Comparison of the Noble Metal Catalysts in the Ethanol Organosolv Lignin Depolymerisation with Hydrogen

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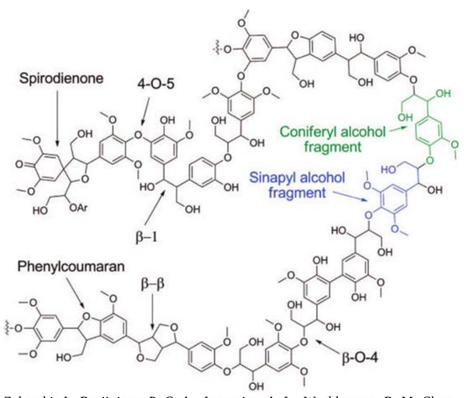
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Introduction

In the bio-based economy, lignin is the most potential resource for aromatic chemicals and fuels, and catalysis is regarded to be the key technology for its valorisation [1]. In addition, hydrogenolysis is considered to be the most promising method to produce phenols from lignin [2]. In this work, various noble metal catalysts were screened in lignin depolymerisation with hydrogen. Ethanol was used as a solvent and a capping agent [3].

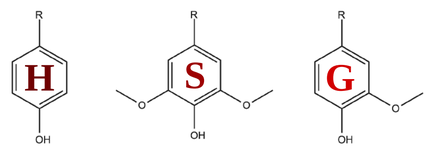
Lignin

- One of the main components of lignocellulosic biomass
  - Binds plant cells together and gives rigidity to the cell walls
  - Structure depends on the source and the separation method
- Significant potential as a source for the sustainable production of fuels, chemicals and materials: unique polyphenolic structure, cheap and renewable, major source for aromatic biocompounds
- Traditionally burned as a low value fuel, only 5% of 50 Mt/a used commercially
- In future, huge amounts of lignin as by-product in the biorefineries



Zakzeski, J., Bruijnincx, P. C. A., Jongerius, A. L., Weckhuysen, B. M. *Chem. Rev.* 2010, *110*, 3552-3599.

H_2 / Catalyst



p-Hydroxy-phenyl units	Syringyl units	Guaiacyl units
1%	**67%**	**32%**

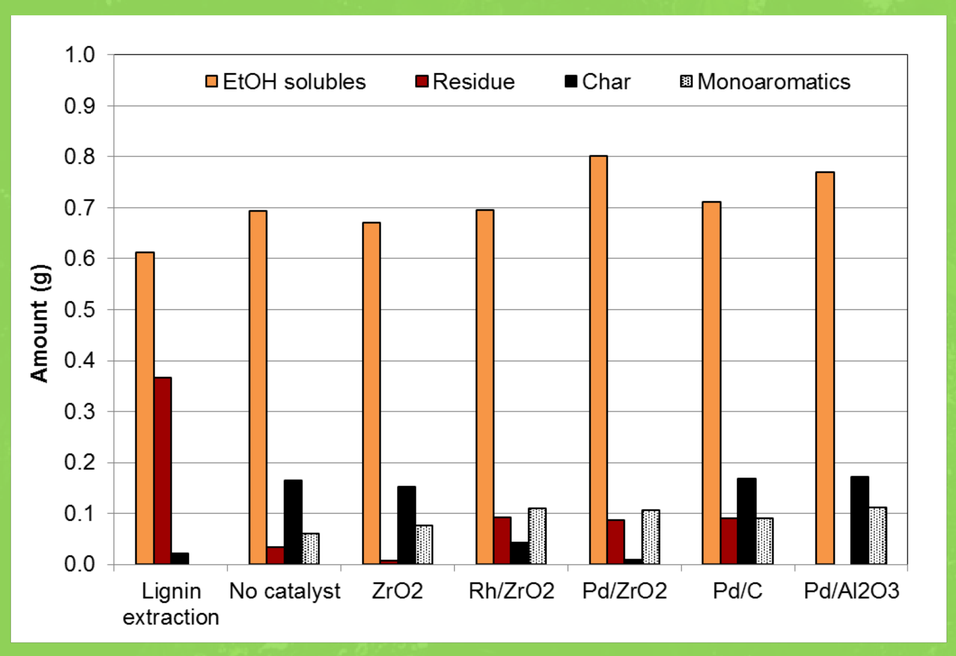
Fraunhofer EOL

Experimental

- 100 ml batch reactor equipped with a spinning basket stirrer (1000 rpm)
- 1 g hardwood (beech) ethanol organosolv lignin (EOL) from the Fraunhofer process in 50 ml EtOH
- 300 °C, 50 bar H_2 (cold pressure), reaction time 1 h
- 100 mg noble metal catalysts with the particle size of 0.30–0.42 mm
  - Commercial catalysts: 3% Pd/C (Degussa) and 5% Pd/Al_2O_3 (Strem Chemicals)
  - ZrO_2 (MEL Chemicals ECO100E) catalysts prepared by impregnation by ourselves
- Analysis methods: GC/MS and pyrolysis-GC/MS, GPC, elemental analysis, GC/TCD-FID (gases), Karl-Fischer titration

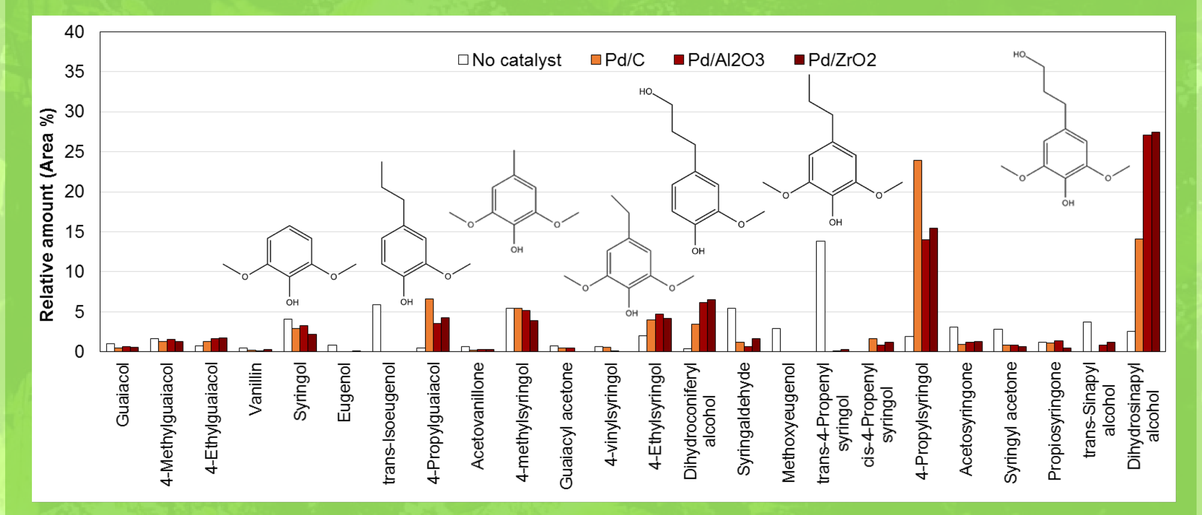
Results

Main Product Fractions



- Catalysts increase the amounts of ethanol solubles and monoaromatic compounds
- ZrO_2 supported noble metal catalysts decrease the amount of char

Monoaromatic Compound Distribution



Effect of the Catalysts

- Amount of monoaromatic compounds increased
- Clearly more hydrogenation and hydrodeoxygenation products
- Rh/ZrO_2: higher amount of products with a shorter side chain compared to Pd/ZrO_2
- Pd/oxide catalysts: less -OH group in propyl side chain compared to Pd/carbon

Conclusions

- Hydrotreatment of EOL on the noble metal catalysts has a large potential in lignin valorisation
- Catalysts have a remarkable effect on the product distributions
- Clear differences between the catalyst metals and supports
- Pd/ZrO_2 catalyst particularly very good, strong metal support interaction assumed
- More catalyst and process development still required

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2. Dorrestijn E.; Laarhoven, L.J.J.; Arends, I.W.C.E.; Mulder, P. *J. Anal. Appl. Pyrolysis* **2000**, *54*, 153-192.
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